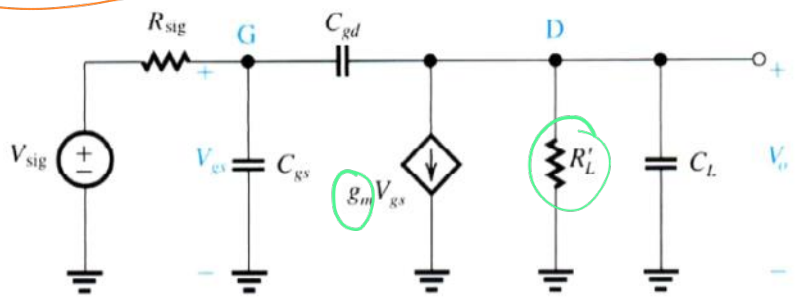
Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.4 High-Frequency Response of Common-Gate and Cascode Amplifiers
- Chapter 10.5 High-Frequency Response of Source and Emitter Fallowers
- Chapter 10.6 High-Frequency Response of Differential Amplifiers

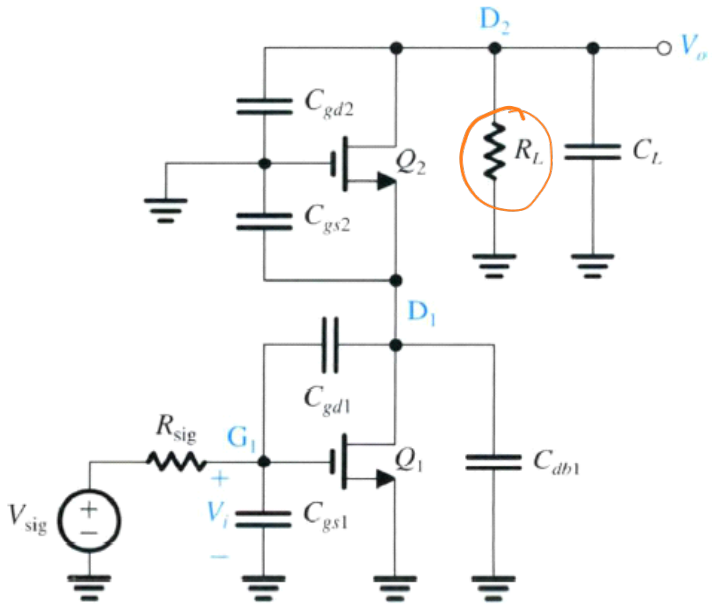
Recap

Common Source Amplifier



$$\tau_H = C_{gs}R_{gs} + C_{gd}R_{gd} + C_L R_{C_L}$$
$$= C_{gs}R_{sig} + C_{gd}[R_{sig}(1 + g_m R'_L) + R'_L] + C_L R'_L$$

Cascode Amplifier



A_o r_o

$R_{d1} = r_{o1} \,//\, \dfrac{r_{o2} + R_L}{g_{m2} r_{o2}}$

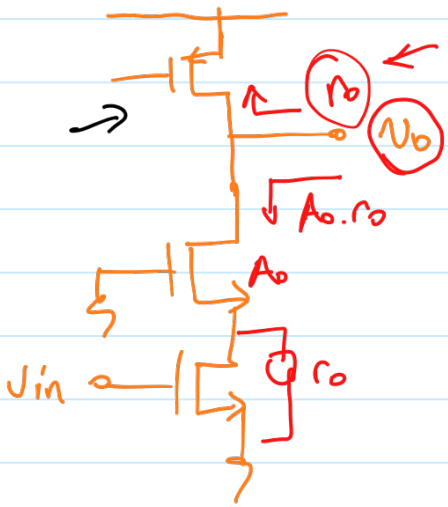
A_o

$$\tau_H = C_{gs1}R_{sig} + C_{gd1}\left[(1 + g_{m1}R_{d1})R_{sig} + R_{d1}\right]$$
$$+ (C_{db1} + C_{gs2})R_{d1} + (C_L + C_{gd2})(R_L \,\|\, R_o)$$

HF Response of Cascode Amp. (High Rsig)

$$\tau_H = R_{sig}\left[C_{gs_1} + C_{gd_1}\left(1 + g_{m_1}R_{d_1}\right)\right] + R_{d_1}\left[C_{gd_1} + C_{db_1} + C_{gs_2}\right] + \left[R_L \| R_o\right]\left[C_L + C_{gd_2}\right]$$

$R_o \rightarrow 1/g_m$

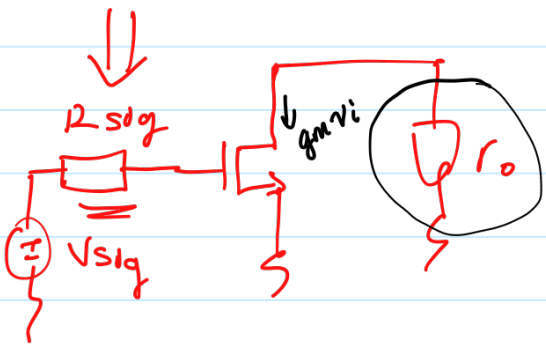


$$R_{d_1} = r_{o_1} \| \frac{r_{o_2} + R_L}{g_{m_2} r_{o_2}}$$

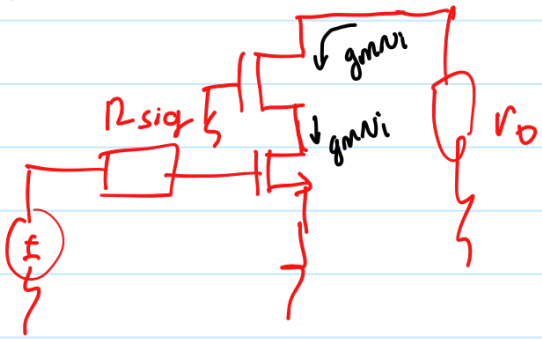
r_o

A_o

$\approx 1/g_m$



$$\tau_H \approx C_{gd} \cdot R_{sig}\left(1 + g_m r_o\right)$$



HF Response of Cascode Amp. (Low Rsig)

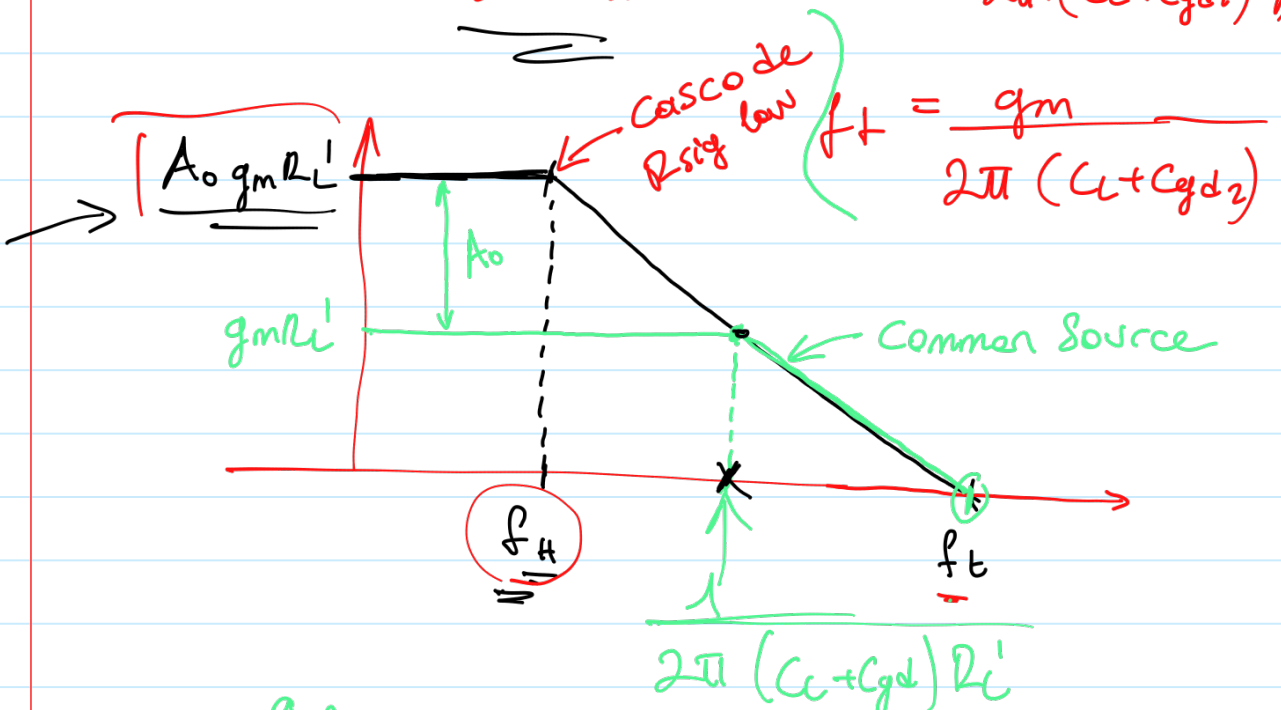
Cascode

$R_L' \approx A_o r_o$

$\tau_H \approx (C_L + C_{gd2})(R_L \parallel R_o)$

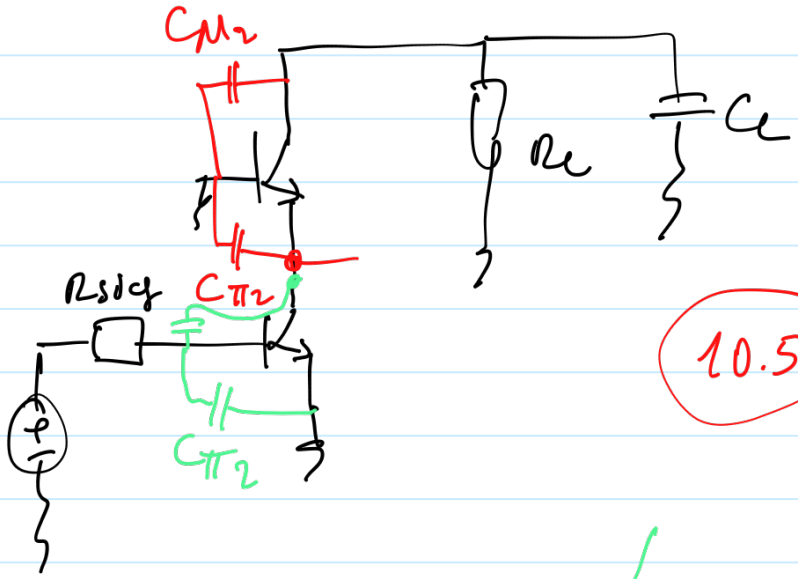
$A_o r_o \quad A_o r_o$

$$f_H = \frac{1}{2\pi \tau_H} \Rightarrow f_t = \frac{A_o g_m R_L'}{2\pi (C_L + C_{gd2}) A_o R_L'}$$



$$f_t = \frac{g_m}{2\pi (C_L + C_{gd})}$$

HF Resp. of the Bipolar Cascode Amp.



10.5

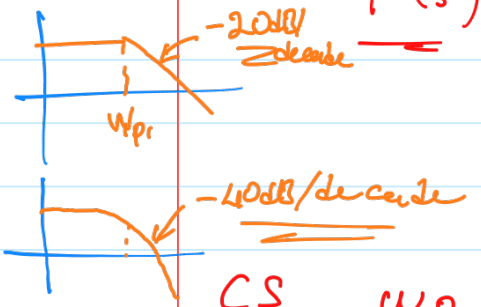
$$\tau_H = C_{\pi 1} R_{\pi 1} + C_{\mu 1} R_{\mu 1} + (C_{cs1} + C_{\pi 2}) R_{c1} + (C_L + C_{cs2} + C_{\mu 2})(R_L \| R_o)$$

$$R_{\mu 1} = R'_{sig}(1 + g_{m1} R_{c1}) + R_{c1}$$

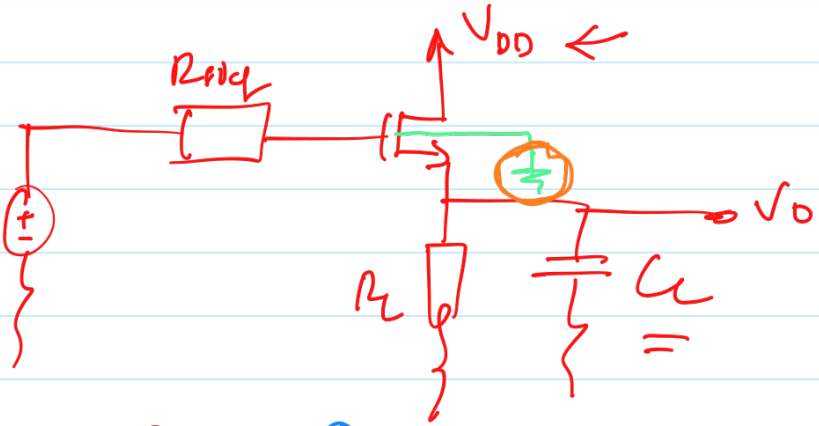
HF Resp. of Source Follower

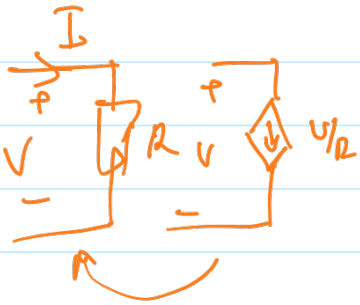


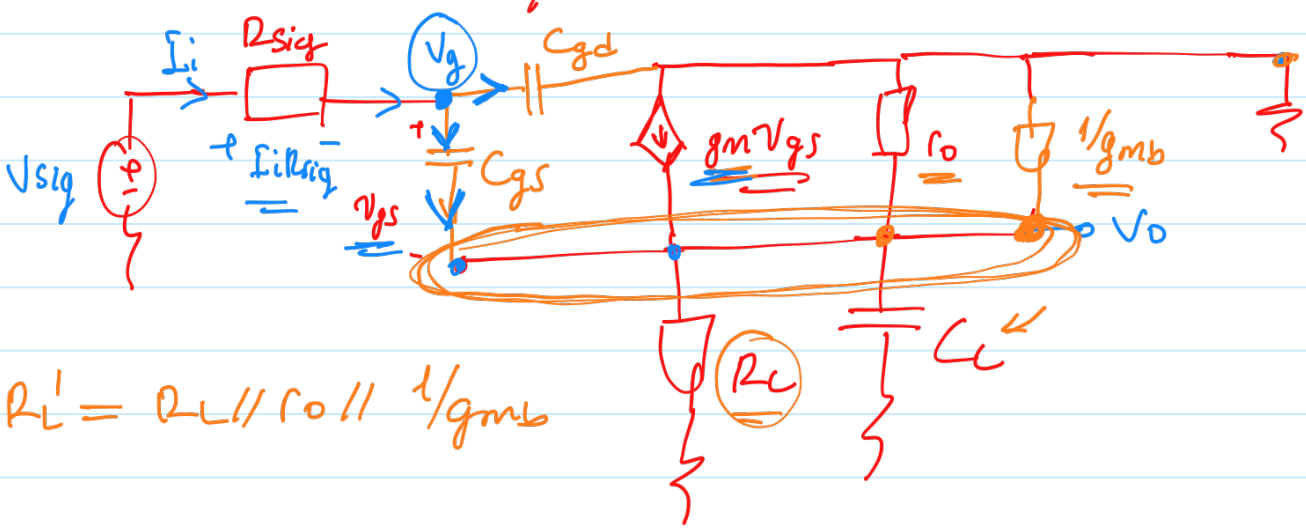
$$F(s) = \frac{\text{- - - -}}{1 + b_1 s + b_2 s^2 \text{ ----}} = \frac{\text{- - - -}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$



CS $\omega_{p2} \gg \omega_{p1}$ ($\omega_{p2} > 4 \times \omega_{p1}$)







$$R_L' = R_L // r_o // 1/g_{mb}$$

$$V_{sig} = I_i R_{sig} + V_{gs} + V_o$$

$$I_i = sC_{gs}V_{gs} + sC_{gd}(V_{gs} + V_o)$$

$$V_{sig} = \left[1 + s(C_{gs}+C_{gd})R_{sig}\right]V_{gs} + \left[1 + sC_{gd}R_{sig}\right]V_o$$

HF Resp. of Source Follower

$$g_m V_{gs} + sC_{gs}V_{gs} = \frac{V_o}{R_L'} + sC_L V_o$$

$$V_{gs} = \frac{1}{g_m R_L'} \frac{1 + sC_L R_L'}{1 + s(C_{gs}/g_m)} V_o$$

$$\frac{V_o}{V_{sig}}(s) = A_M \frac{1 + \frac{s}{\omega_z}}{1 + b_1 s + b_2 s^2}, \quad \omega_z = g_m / C_{gs}$$

$$A_M = \frac{R_L'}{R_i' + 1/g_m}$$

$$b_1 = \left(C_{gd} + \frac{C_{gs}}{g_m R_L' + 1}\right) R_{sig} + \left(\frac{C_{gs} + C_L}{g_m R_L' + 1}\right) R_L'$$

$$b_2 = \frac{(C_{gs} + C_{gd}) C_L + C_{gs} C_{gd} R_{sig} R_L'}{g_m R_L' + 1}$$

$$1 + b_1 s + b_2 s^2 = \left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)$$

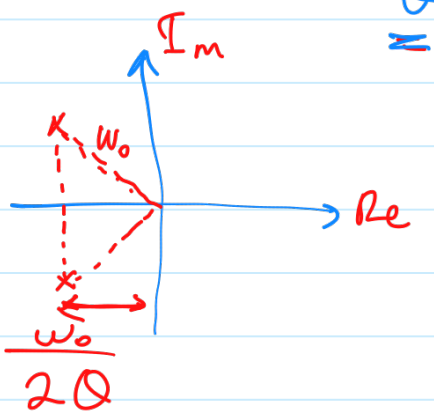
$\omega_{p2} \gg \omega_{p1}$ $\quad f_H = \frac{1}{2\pi b_1}$

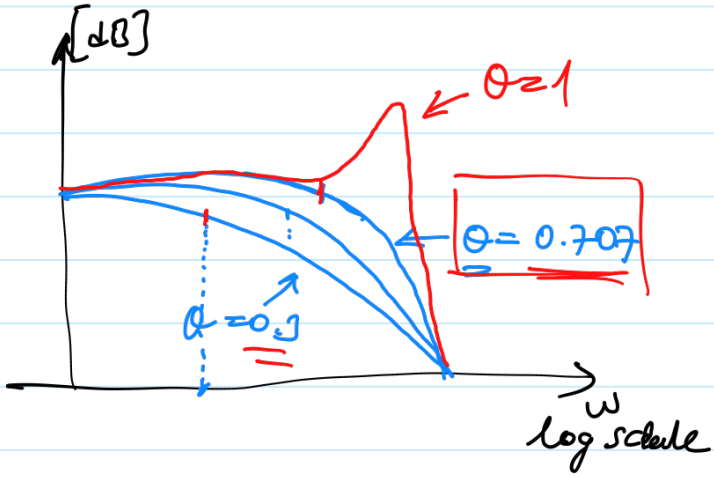
HF Resp. of Emitter Follower

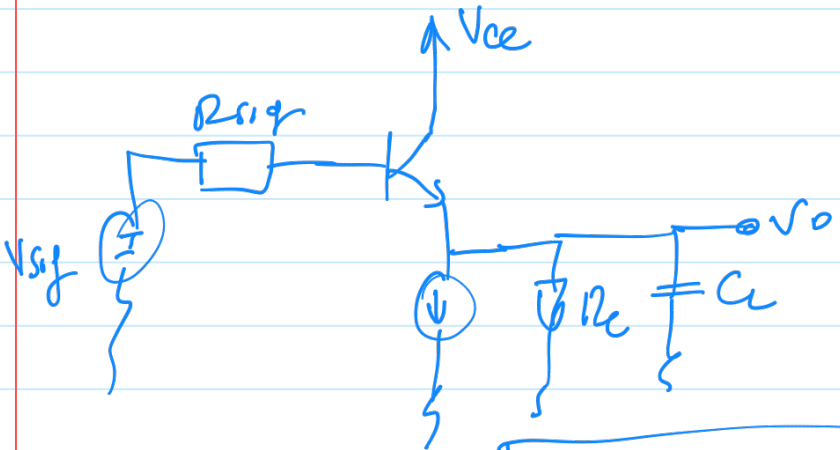
$$f_H \approx \frac{1}{\sqrt{\frac{1}{f_{P_1}^2} + \frac{1}{f_{P_2}^2} - \frac{2}{f_Z^2}}}$$

$$1 + b_1 s + b_2 s^2 = 1 + \frac{1}{Q}\frac{s}{\omega_0} + \frac{s^2}{\omega_0^2}$$

$Q > 0.5$

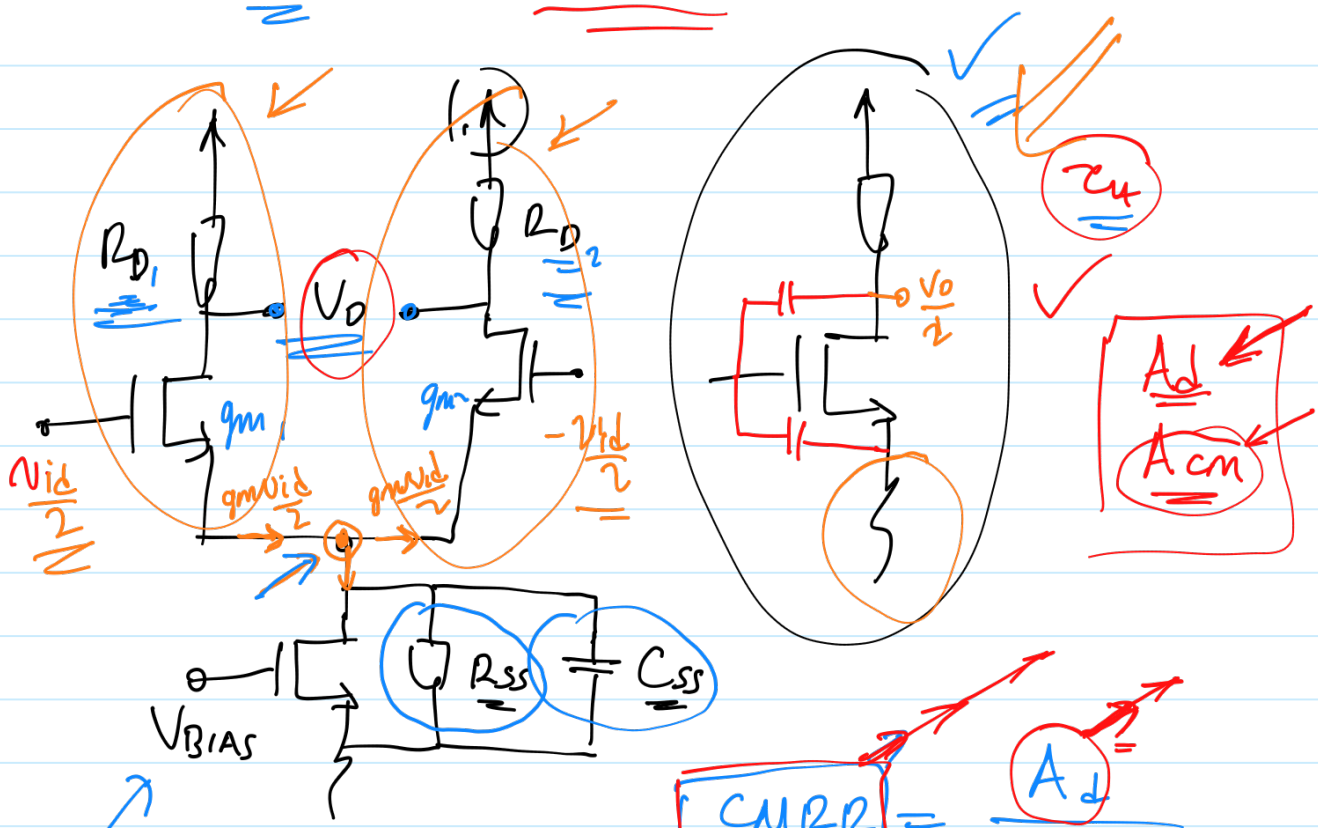




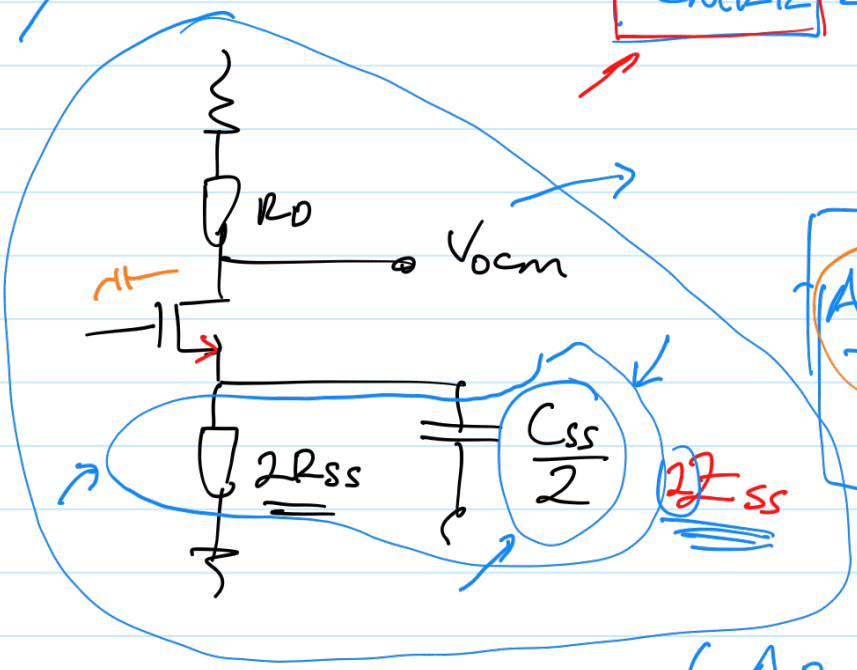


Chapter 10.5

HF Resp. of Res. Loaded Differential Amp.



$$CMRR = \frac{A_d}{A_{cm}}$$



$$A_{cm} = -\left(\frac{R_D}{2R_{ss}}\right)\cdot\frac{\Delta R_D}{R_D}$$

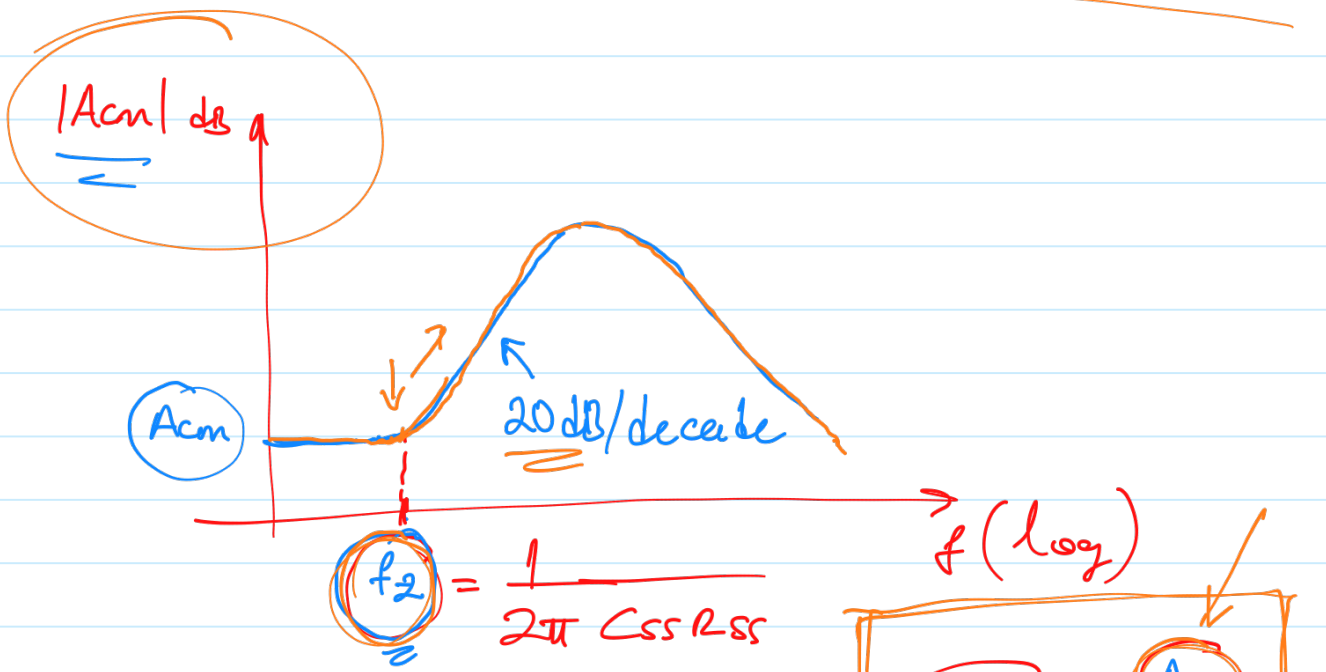
$$A_{cm}(s) = -\frac{R_D}{2}\cdot\left(\frac{\Delta R_D}{R_D}\right)Y_{ss}$$

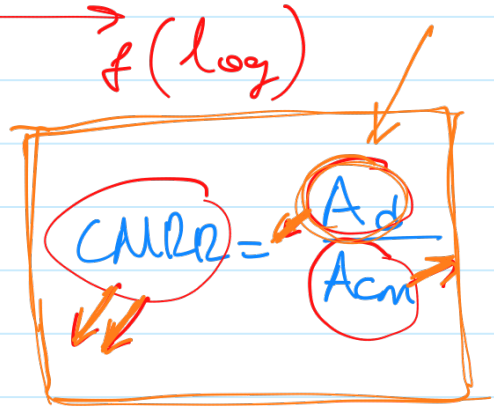
$$A_{cm}(s) = -\left(\frac{R_D}{2R_{ss}}\right)\left(\frac{\Delta R_D}{R_D}\right)\left(1 + sC_{ss}R_{ss}\right)$$

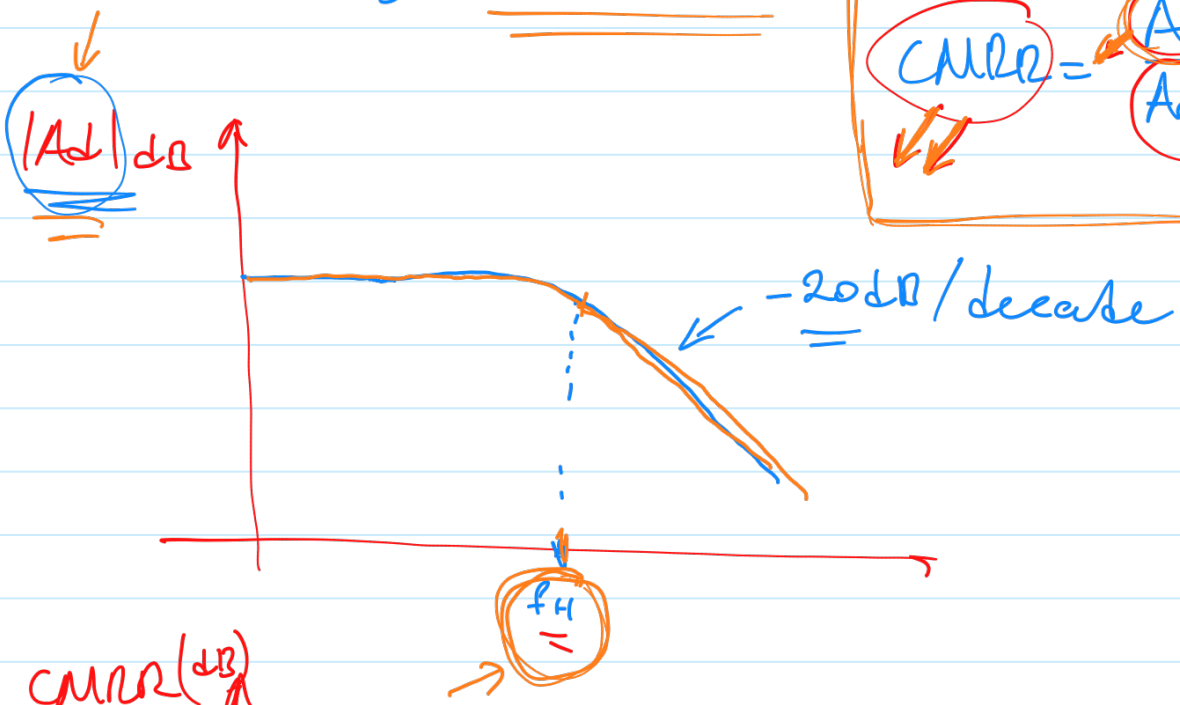
A_{cm}

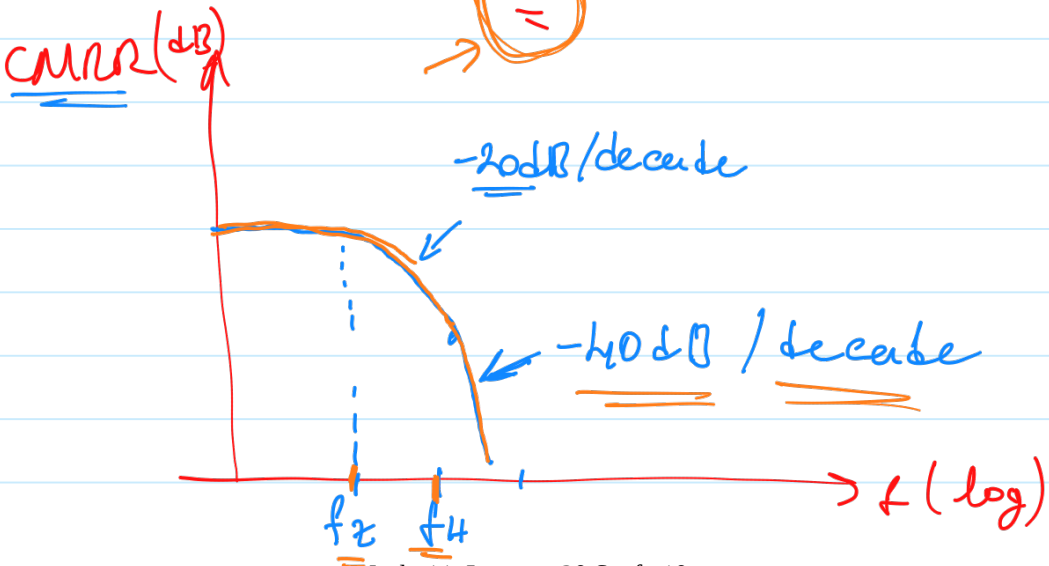
HF Resp. of Res. Loaded Differential Amp.

$$\omega_z = \frac{1}{C_{SS} R_{SS}} \qquad f_z = \frac{\omega_z}{2\pi} = \frac{1}{2\pi C_{SS} R_{SS}}$$









HF Resp. of CM Loaded Differential Amp.

HF Resp. of CM Loaded Differential Amp.

Template

Template

Template

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